

COVID-19 Pandemic and Sustainable Development Goals in Nigeria: Focus on Blended Teaching and Learning

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Abstract

The global catastrophe occasioned by the COVID-19 pandemic in 2020 disrupted the whole economic sector in Nigeria and threatened both the educational system and Sustainable Development Goals 4 (SDG 4). In response to the pandemic, Nigerian schools were closed from March 23 till September 2020, to prevent the spread of the virus, leading to a halt in traditional face-to-face education. While some private schools shifted to virtual learning, public schools were unable to do so due to a lack of necessary infrastructure, highlighting deep disparities in the education system. Therefore, this study aimed to examine the role of blended learning in sustaining education in the face of a global disaster with a view to meeting SDG 4. The methodology was qualitative with the use of primary data such as in-depth interview, key informant interview and data descriptive method that would rely on secondary sources. The researchers identified a critical gap in Nigeria's preparedness for educational continuity in times of crisis, particularly for public schools, and proposed blended learning as a viable

solution to bridge this gap. Findings suggest that for Nigeria to achieve SDG4, a resilient approach that integrates both physical and virtual learning methods is essential to ensure education continuity in the event of future global disruptions. This blended approach will be crucial to achieving equitable education and promoting lifelong learning opportunities for all.

Keywords: COVID- 19 pandemic, Blended learning, Education, Sustainable Development Goals, Nigeria

Introduction

The global catastrophe occasioned by the COVID-19 pandemic in 2020 disrupted Nigeria's economic sector, threatening both the educational system and Sustainable Development Goals 4 (SDG 4), which is to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all. COVID-19 affected all sectors of education in Nigeria and international communities. Commencing from March 2019, the global world, including Nigeria, was brought to a standstill due to the outbreak of a novel deadly disease caused by the coronavirus, popularly known as COVID-19. This pandemic disrupted educational programme in March 2020 and drastically affected the teaching and learning mode in the country. Schools in most countries of the world were closed to protect students from the disease (Nigeria Centre for Disease Control (2020). As a preventive measure, the Ministry of Education, following the directive of the Federal Government, ordered the closure of all levels of education (from pre-primary to tertiary) in Nigeria, with effect from 23rd March 2020, to curb the spread of the disease. This development was of great significance to SDG 4. The traditional face-to-face mode of learning was suspended; public schools in the country could not operate virtual learning due to lack of infrastructure to do so. Although some schools in some countries resorted to virtual/online or e-learning for teaching and learning, others did not find it cost-effective, especially in Nigeria, where parents had been facing challenges sponsoring children in schools (Eli-Chukwu et al., 2023; Obaleye et al., 2022; Agbele & Oyelade, 2020). This development was an eye opener for the educational sector to adopt a blended learning mode that would be responsive to continuity in education for sustainable development.

Blended learning, which is the integration of technologies into education, is trending now in the global educational system (Thahir, Widiawati & Baitillah, 2023; Adeoye, Adanikin & Adanikin, 2020). It has its roles, benefits, and challenges. It is regarded as a learning mode that is responsive to continuity in education. Blended teaching and learning, as often used, refers to the use of new forms of technological tools that involve interactive participation online, synchronous and asynchronous, to support teaching and learning in a face-to-face traditional classroom (Williams, 2024; Bizami, Tasir & Kew, 2023; Li, et al., 2022). The blended learning mode is recognized as an important tool for educational advancement in the era of globalization. This is because it functions both face-to-face and virtually. Integrating technologies in education will bring new trends in teaching, networking, and e-learning among teachers and learners. To achieve SDG 4, educational systems must adopt a teaching and learning system that will ensure continuity in education; that will meet learners' needs and sustain education in the event of a pandemic, as in the case of Ebola and COVID-19.

The blended learning method is capable of sustaining education during periods of disaster since it is flexible and customized to students' expectations. With this system, the suspension of education in public schools will be a thing of the past globally. It therefore means that during this new normal in the world, failure to embrace blended learning would continue to lead to the suspension of education in public schools. This suspension signals danger for educational systems. Hence the justification for the global shift to embrace digital innovation and technology along with traditional face-to-face teaching and learning for innovation in the education sector, innovation for sustainability in education and invariably to promote the attainment of the sustainable development goals (Moses, Wugate & Nlensee, 2023; Vaughter & Yamaguchi, 2023; Kayode & Lingaliso, 2023)

Literature Review

Coronavirus infection: A conceptual analysis

COVID-19 has been defined in different ways. The World Health Organization (WHO) (2020) defined it as a highly transmittable and pathogenic viral infection. According to Kente (2020), the WHO also

described the virus as "zoonotic" which means that the disease is transmitted between people and animals and an illness caused by a novel coronavirus now called severe acute respiratory syndrome coronavirus 2 (SAR COV-2). Researchers have carried out studies on the COVID-19 pandemic. Some studies found that COVID-19 is a disease that has spread across a large region and multiple continents in the world (Agbele & Oyelade, 2020). Ekpenyong (2020) explained how the whole world was brought to a standstill late in 2019, due to the outbreak of a novel deadly disease called COVID-19. Kente asserted that the first confirmed case of coronavirus disease in Nigeria was announced on 27th February 2019, when an Italian citizen tested positive for the virus in Lagos. He also revealed that the second case of the virus was reported in Ewekoro, Ogun State, where a Nigerian citizen came in contact with an Italian citizen (Nigerian Tribune, cited in Kente 2020). In addition, reports, by the NCDC (2020) gave a case definition of COVID-19 infection as any person with acute respiratory illness (fever and either cough, difficulty breathing, or shortness of breath). The disease mainly spreads through respiratory droplets during face-to-face close contact with an infected person. The infection was also observed to be spread by asymptomatic, pre-symptomatic, and symptomatic carriers (NCDC, 2020). Again, the NCDC (2020) recorded the symptoms exhibited by COVID-19 infection as breathing difficulties, shortness of breath, fever, respiratory problems, and cough. The impact of COVID-19 on education is bizarre and unprecedented in history as about 1.5 billion students from across the globe were affected by school closures.

Effect of COVID-19 Pandemic on Nigerian Educational Sector

The effect of the COVID-19 pandemic on the Nigerian educational sector was very severe and a threat to the attainment of SDG 4, which is to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all. The pandemic led to a partial discontinuity in education in the country because schools were forced to shut down during the lockdown period. After the historic disruption by the COVID-19 pandemic, private schools adopted virtual learning while public schools could not (Agbele & Oyelade, 2020). The closure of schools at all levels drastically threatened the attainment of SDG 4.

All sectors of the economy were brought to a standstill but the educational sector was most affected. This development was a clarion

call for the educational sector to consider the blended learning mode for continuity. During the post-COVID-19 era, neither the traditional face-to-face nor the virtual method could sustain education in the face of global disaster. Therefore to sustain the educational sector, blended learning should be considered for the attainment of SDG 4.

Sustainable Development (Goals (SDGs): A Concept

The concept of sustainability is the ability to maintain or support a process over time. Sustainable development is the idea that human societies must live and meet their needs without compromising the ability of future generations to meet their own needs. This study asserts that there is a relationship between education and sustainable development since education promotes the development of knowledge, skills, understanding, and values and actions required to create a sustainable world. The official definition of sustainable development was first developed in the Brundtland Report in 1987. The study found that the Sustainable Development Goals aim to transform the world and are a call to action to end poverty and inequality, protect the planet, and ensure that all people enjoy health, justice, and prosperity (UNDP, WHO & FAO of the UN 2020). This has made it critical to the extent that no one should be left behind in the process of education. Research revealed that in 2015, all the countries in the United Nations adopted the 2030 agenda for Sustainable Development with 17 goals.

According to the United Nations, "the 17 SDGs are integrated – they recognize that action in one area will affect outcomes in others and that development must balance social, economic and environmental sustainability". Out of the 17 Sustainable Development Goals, No 4 is devoted to quality education, and the aim is to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all (FAO of the UN). Education will allow every human being to acquire the knowledge, skills, attitude, and values necessary to shape a sustainable future. In line with this, Vladimirova and Blanc (2015) identified a link between education and the sustainable development goals and emphasized blended teaching and learning to motivate and sustain learning (Yadav, 2016). Vladimirova & Blanc (2015) also gave high priority to education in the development goals agenda. Therefore, continuity in education is the key to the achievement of the sustainable

development goals so that educated people can break the shackles of poverty.

Concept of Blended Learning and Sustainable Development

The concept of blended learning arose in Nigeria in 2020, when the COVID-19 pandemic disrupted the whole economic sector, threatened the educational system and SDG 4. Researchers have defined blended learning in a variety of ways. Doubet and Carbaugh (2020) explained the components of blended learning as launch/hook, face-to-face, introducing new contents of skills; formative assessment and checklist for understanding; active processing with peers and authentic learning experience. Kayode and Obiseye (2022), Rasheed, Kamsin and Abdullah (2020), and Kayode (2019) defined blended learning as an approach that combines the benefits afforded by face-to-face and online learning components. A study revealed that blended learning is also an approach that mixes structured classroom trends with online content in the form of video (Biewener, 2022; Govender & Kayode, 2020). This means that the blended learning approach, also known as hybrid learning, is capable of sustaining education in the face of any uncertainty, thereby ensuring inclusive and equitable quality education and promoting lifelong learning opportunities for all.

Role of Blended Learning and Digital Innovation in Future Disasters

The role of blended learning in sustaining education in the face of global disaster cannot be over-emphasized. Blended learning is more critical during a prevailing period global pandemic. After the COVID-19 pandemic in Nigeria, blended learning was used to close the gap in educational needs of Nigerians. The adoption of blended learning in some institutions enabled teachers to share instructional materials with learners through the use of digital tools. Studies reveal that the use of digital tools gives students room to learn in their own time using networking tools to collaborate and interact in learning activities among themselves.

Researchers also stress that blending the face-to-face traditional method with technological tools is what would work globally in teaching and learning, and must be enforced. (Kayode, 2019). The Nigerian educational sector must adopt the blended learning mode as the best educational approach to forestall closure of schools during

global disasters. This method allows students to learn and progress at their own pace and adapt to using some of these technological tools in their daily lives. Therefore, stakeholders in the educational system must be more proactive and committed in the use of remote and digital tools to avoid disruption in education in case of a global disaster such as COVID-19.

Benefits of Blended Learning to Students in the Face of Future School Lockdown

The benefits of blended learning are relevant in the educational sector in this era of uncertainty about natural disasters in Nigeria as the country must meet the goal of sustainable learning of SDG 4. For some decades now, technology has brought significant changes to the education sector globally. The blended learning method is beneficial to education because of its flexibility as it could be integrated into any classroom anywhere, anytime. One of the benefits of this method is that learners are in control of their learning pace. It also saves money and is modular and scalable (Biewener, 2022). Research has found it to be very helpful in personalized education for individual students (Weitzel, 2021). From experience, both learners and teachers benefit from blended learning because they all operate in a technology-driven world. Studies have revealed that blended learning engenders learners to acquire technology that will give them access to computers and mobile devices at their convenience (Weitzel, 2021; Biewener, 2022).

This researcher observed that, during the post-COVID-19 era, some private universities in Nigeria were using mobile access, tablets, and smart phones. Teachers were able to schedule assignments, score students, and track students' progress through online facilities. Researchers have found that the use of Information and Communication Technologies (ICT) or digital tools has had a positive influence on the educational environment (Kayode et al., 2019). Also previous researchers have supported the use of technological tools to enhance teaching and learning (Chin et al., 2019; Caird & Roy, 2018). Mtebe and Raphael (2013) noted that the University of Dar es Salaam, has witnessed increased student enrolment through blended distance learning.

Challenges of Effective Implementation of Blended Learning in Nigeria

Studies have revealed that numerous challenges affect the effective implementation of blended learning in Nigeria. Some of them include: expensive cost of web and software development and maintenance; high cost of blended learning infrastructure; nature of Nigeria's school curriculum; lack of infrastructure and amenities; persistent epileptic power supply; unequal access to technology usage by students; inadequate capacity building for both teachers and students; lack of confidence in blended learning (Neel, 2019). Rasheed, Kamsin and Abdullah (2020) and Mtebe and Raphael (2013) also found that outdated learning resources; underutilization of learning centres, and technical difficulties are some of the main challenges. A study by Olelewe (2014) revealed some of the major challenges facing the application of blended learning in teacher education as: finding the right blend, increased demand on time, overcoming barriers of institutional culture; inadequate infrastructure, limited access to internet facilities, and insufficient allocation of funds to the educational sector.

Challenges of Blended Learning for Sustainable Development

In order to meet Sustainable Development Goal 4, provision of adequate basic technological tools for the smooth running of blended learning by the federal government is required. Blended learning comes with challenges such as provision of space for learning, personal distraction-free space; balancing the use of gadgets inability of teachers and students to stay connected; inadequate digital tools and collaboration (Olelewe, 2014); and financing and maintenance of infrastructures in teaching and learning. Lack of partnership with international donor agencies and private sector investors to train personnel and provide infrastructure is also a challenge.

Statement of the Problem

During the outbreak of COVID-19, both the country and schools from pre-primary to tertiary levels were shut down. The closure of schools drastically affected the teaching and learning mode in the country. Public schools in the country could not operate virtual learning due to lack of infrastructure. The Federal Government of Nigeria and the Nigeria Centre for Disease Control and Protection (NCDC) were at the

forefront to combat COVID-19 infection. During the post-COVID-19 era, it was discovered that neither the traditional face-to-face nor the virtual method of teaching and learning would meet Nigeria's educational needs in the face of a global disaster. The absence of a blended learning method that would enable continuity in education would be translated to a bottleneck in the educational sector as well as a negative effect on the attainment of SDG 4. As a result of this realization, the researchers found a gap in future education in Nigeria and are therefore seeking to close this gap by proposing blended learning as a learning mode that is responsive to continuity in education in the country, to put an end to future schools closure in the face of an impending disaster. It is against this backdrop that this paper seeks to answer the following research questions: What is the role of blended learning in sustaining education at all levels in the face of a global disaster in Nigeria? What is the benefit of blended learning to students in case of another school lockdown? What are some of the constraints to effective implementation of blended learning in Nigeria and what are the immediate solutions to these constraints?

Research Objectives

The main objective of this study is to examine the role of blended learning in sustaining education at all levels in the face of a global disaster in Nigeria. The specific objectives are to:

1. Highlight the benefits of blended learning to students in case of another school lockdown.
2. Determine the constraints to effective implementation of blended learning in Nigeria and the immediate solutions to these constraints.

Research Questions

The research therefore, raised the following questions to guide the study:

1. What are the benefits of blended learning to students in the face of another school lockdown?

2. What are some of the constraints to the effective implementation of blended learning in Nigeria and the immediate solutions to these constraints.

Ibadan Metropolis

Ibadan metropolis is the location of the study. It is the capital of Oyo State in southwestern Nigeria and is the largest urban centre in the state. Its elevation is approximately 230 metres (755ft) above sea level. It is in the tropical savanna climate zone with two rainy seasons. It is also the major commercial centre for the agriculture, manufacturing, construction and service sectors. Ibadan is the largest city in West Africa by land area with a population of about 3.2 million. The predominant ethnic group in the metropolis is Yoruba. Ibadan has a very rich cultural heritage and is home to University of Ibadan. Ibadan boasts a National museum, Mapo Hall, Ibadan Zoo, the Olubadan Palace and a well-connected road network.

Methodology

Ibadan metropolis has a total of 133 secondary schools spread across five local government areas. Five secondary schools in Ibadan North LGA were purposely selected due to their use of blended teaching technology. The five schools selected constituted 3.76% of the schools in the LGA. The selected schools met the specific objectives of the study because they possess unique features and resources relevant to the goals of the project. A descriptive survey design was adopted for the qualitative analysis of this study. The population comprised the teachers and students in the five selected secondary schools. The convenience sampling procedure was used to select teachers who were willing to participate in the study, while the purposive sampling technique was deployed to select brilliant students by the teachers. The total number of respondents for the study was 100; a total of 50 teachers and 50 students were required from each of the five schools. Thus, 10 interviewees were selected from each of the schools.

In-depth interviews (IDI) were arranged and held with the teachers while key informant interviews (KII) were held with the students. The purposive sampling technique was adopted in selecting the teachers in order to deliberately recruit experienced teachers and students who were identified by teachers as brilliant, and to give equal

opportunity and representation to both teachers and students. The descriptive survey was adopted to explore secondary sources of data such as the internet, documents from libraries, manuscripts, textbooks and journal articles. The essence of descriptive research is to enable the researchers collect data from a representative sample and generalize the result relative to the topic of the study. Other data were generated from the current related literature on blended learning and its relevance to education and sustainable development. The study employed simple deductive reasoning as a method of data analysis based on the importance and validity of available data.

Findings and Discussion

Based on the responses from the interviewees, thematic analysis of the data revealed that there is a gap between face-to-face and virtual learning, and the adoption of blended learning would fill this gap in case of an outbreak of a global plague or a pandemic. Responses from interviews revealed that blended learning is relevant to sustain the future of education, even in the event of a global pandemic and lockdown. It was also revealed that this method would enable instructors to cover their course content within a specified period, even during a pandemic lockdown period; promote the production of knowledge and critical thinking; enhance constructive communication; promote interactive learning between teachers and learners; enhance collaborative learning in classrooms; and students would become life-long learners. It was found that it would play a vital role in sustaining education in the face of global disaster at all levels of education and would meet the teaching and learning needs of Nigeria.

The study also revealed that teachers, lecturers, parents, and stakeholders stand to benefit from the use of blended learning. It also showed that students in Nigeria are ready to yield to the new changing mode of learning during this digital age. In addition, it was found that blended learning would create more opportunities for students to explore more technological tools with blended methods together, synchronously or asynchronously, at their own pace. The study found that the integration of digital technology in teaching and learning cancel out the barriers to education during a lockdown in the case of any disaster/pandemic. There would be continuity in teaching and learning for teachers and students.

The study also revealed one of the benefits of blended learning as an increase in the usage of different digital tools and applications among students. Teachers and students confirmed that continuity in teaching and learning will promote the achievement of the sustainable development goals.

On the other hand, the results also revealed some challenges, such as epileptic and unsteady power supply; lack of infrastructure and basic amenities; high cost of implementation of blended learning tools; nature of Nigerian school curriculum; inadequate capacity training for both teachers and students, and so on. Findings also revealed suggestions of how some of these challenges could be mitigated, including: the Federal Government of Nigeria must find immediate solutions to challenges of power supply currently plaguing the country; funding basic amenities and educational infrastructure; and payment of student's bursaries to meet expenses on data. These steps would enhance the educational system and make education survive any pandemic situation. Thus, blending the learning method is a panacea to continuity in education and attainment of SDG 4.

Conclusion

Continuity in the Nigerian educational system in the 21st century will, to a large extent, depend on the method of teaching and learning. The COVID-19 pandemic has shown that a pandemic can arise at any time and inflict negative impacts on the Nigerian educational system. This study has shown that blending both face-to-face and virtual methods of learning will afford Nigeria the opportunity of sustaining education at all levels if faced with a global disaster. Blended learning has proven to be more relevant to continuity in teaching and learning in periods of unexpected national or global disaster than online or face-to-face learning alone. The blended learning method would be beneficial to the Nigerian educational system because it is flexible; it can be integrated into almost any classroom anywhere, anytime; learners are in control of their learning pace, and it saves money. The researchers found that blended learning favours both teachers and learners because this mode of learning allows students to progress in their own space. This process of effective usage of digital tools with the traditional method will enhance and improve teaching and learning. The use of innovative technology to complement the traditional face-to-face method of teaching and learning

will enhance the continuity of education and promote achievement of the sustainable development goals. High quality of education and equal access to education for men and women could be achieved through the blended learning mode. Blended learning has the potential to solve the problem of access to education in the event of any future disaster or insurgency in the country.

The challenges of effective implementation of blended learning in Nigeria which include: epileptic power supply, high cost of data, unequal access to technology, unsteady internet access, lack of infrastructure and basic amenities, high cost of implementation of blended learning tools; nature of Nigerian school curriculum; and inadequate capacity training of both teachers and students must be addressed to successfully implement blended learning in case of a global epidemic shut down. Education is very important and must not be disrupted because all the knowledge acquired through education is utilized for the attainment of sustainable development. Thus, it is imperative to adopt blending learning to meet the teaching and learning needs of the country as well as sustainable development goals.

Recommendation

This study recommends that the Federal Government should fashion out a policy to fuse blended methods of teaching and learning into the mainstream of the Nigerian Educational Curriculum. This will enhance and offer the kind of education the country requires to meet SDG 4. Nigeria's government should equally consider an increment in budgetary allocation for all levels of education in the country; inject funds into electricity generation and overhaul the national grid for constant supply of electricity. Teachers should also develop the ICT pedagogy model that will improve virtual teaching and learning synchronously or asynchronously as an innovative method during this new normal world.

Conflict of Interest

The authors declare that there is no conflict of interest allied with this research work or its findings.

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